

CS 548: Cloud Networking

Course Code	CS 548
Course Name	Cloud Networking
L-T-P-C	3-0-0-3
Prerequisites	CS 406 Computer Networks, CS 310 Introduction to Computing Distributed Processes, CS5 47 Network Management Systems, or the instructor's consent
Intended for	UG/PG
Distribution	Elective
Approval	9th Senate (May 2016)

Course Contents

- **Module 1:** Cloud Networking Introduction. [3 Lectures]
- **Module 2:** [7 Lectures]
 - Application and traffic patterns - how web search works, data center traffic , implication on networking
 - Physical network structure - Big Switch Approach, FAT tree network etc
- **Module 3:** [12 Lectures]
 - Routing and traffic engineering - STP(spanning tree), link stat protocol, Transparent Interconnection of Lots of Links, OSPF over IP, Border Gateway Protocol in the Data Center, distributed congestion aware load balancing for data centers.
 - Host virtualization - Server Virtualization, networking VMs, Improving networking performance, packet processing on CPUs, Open vSwitch.
 - Congestion control - Feedback control loop, Basic Congestion control Loop, Problems with TCP, Data Center TCP, Explicit Congestion Notification.
- **Module 4:** [8 Lectures]
 - Introduction to SDN architecture - Software-defined Networks, evaluation of SDN, flexible data planes, logically centralized control.
 - Network virtualization - Network virtualization in VL2, Network virtualization in VMware NSX (VL2 Physical topology, Routing in VL2, Routing Implementation, end-end example).
- **Module 5:** [10 Lectures]
 - Inter-data center networking- Large online services work, Traditional WAN approach and problems.
 - Content Distributed Networks
 - Application layer techniques
- **Module 6:** Green Cloud Networking. Energy efficient Optimization Technology, Nano Data Center. [3 Lectures]

Text Books

1. G Varghese, **Network Algorithmics**, 1st Edition An Interdisciplinary Approach to Designing Fast Networked Devices, 2014
2. Lee Chao, **Cloud Computing Networking: Theory, Practice, and Development**, CRC Press, August 2015.
3. William Stallings, **Foundations of Modern Networking: SDN, NFV, QoE, IoT, and Cloud**, Addison Wesley, Oct 2015.
4. Ken Gray, Thomas Nadeau, **SDN - Software Defined Networks**, O'Reilly Media, August, 2013.

References

There will be a selection of readings and resources available to you for this course. Some of these will be required, while others will be optional for you to explore further.

1. Use of BGP for Routing in Large-Scale Data Centers, online <https://www.ietf.org/id/draft-ietf-rtgwg-bgp-routing-large-dc-07.txt>
2. RFC, TCP Congestion Control (This is an RFC, i.e., Request for Comments.). (online) <https://www.ietf.org/id/draft-ietf-rtgwg-bgp-routing-large-dc-07.txt>, <https://tools.ietf.org/html/rfc5681>
3. Mohammad Alizadeh et. al., **CONGA: Distributed Congestion-Aware Load Balancing for Datacenters**, SIGCOMM14, August 17-22, 2014, Chicago, IL, USA.
4. Ben Pfaff et. al., **The Design and Implementation of Open vSwitch**, 12th USENIX Symposium on Networked Systems Design and Implementation (NSDI 15).May 4-6, 2015, Oakland, CA, USA
5. Ankit Singla et.al, **Towards a Speed of Light Internet**, ACM Hotnets, Oct. 2014, LA, USA 2014
6. Ilya Grigorik, **Latency: The New Web Performance Bottleneck**, 2012.